Gitesh Kumar

AI Engineer | Industrial Systems & Generative AI

Machine learning engineer with experience building AI-driven systems that combine structured data pipelines with modern generative AI models. My recent work focuses on designing practical LLM workflows and agent-based systems that translate raw operational data into decision-ready insights. With a background in engineering and 4+ years experience working with industrial data systems in the energy sector, I enjoy exploring how ML and Gen-AI can improve real engineering processes.

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EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg — From October 2022 to March 2026
Masters in Elektromobilität-ACES
Specialization: AI and Sustainable Production Technology

PEC University of Technology Chandigarh, India — From July 2013 to May 2017
Bachelors of Engineering
Mechanical Engineering

RECENT ACADEMIC PROJECTS

Generative AI Agent for Industrial Engineering

- Designed a Medallion data pipeline to transform raw factory sensor data into structured operational KPIs.
- Developed a LangChain-based ReAct agent using a locally hosted Llama 3 model to autonomously generate SQL queries against engineering datasets.
- Built a privacy-first local system allowing engineers to query sensitive operational data without sending information to external AI services.
- Demonstrated how AI-assisted analytics can support operational decision-making without exposing sensitive data to external cloud APIs.
- Designed for deployment in safety-critical industrial environments where data privacy and operational reliability are paramount.

Master's Thesis: A Retrieval-Augmented Framework for Fine-tuned LLMs for Vulnerability Analysis

- Developed and evaluated a retrieval-augmented LLM pipeline for cybersecurity analysis using LLaMA-based models and domain-specific datasets.
- Analyzed strengths, limitations and failure modes of domain-adapted LLMs in vulnerability and cybersecurity reasoning.
- The focus was on de-risking the technology by conducting systematic stress tests, proving that we can make advanced AI reliable enough for a global enterprise.
- Designed controlled experiments comparing baseline and RAG-enhanced models using NLP evaluation metrics and statistical testing.

WORK EXPERIENCE

Sr. Operations Officer — From July 2017 to August 2021
Hindustan Petroleum Corporation Limited India

- Worked with plant engineering teams to integrate operational sensor data and ERP data into structured datasets supporting performance monitoring and production analysis.
- Designed data workflows combining SCADA data, operational logs, and ERP system to improve visibility into plant performance.
- Acted as the main point of contact between business users and technical service providers like Honeywell to ensure automation projects met operational needs.
- Developed reporting tools that allowed engineering teams to interpret machine performance data and identify operational inefficiencies.
- Delivered a 10 percent increase in production efficiency through data driven decision making by streamlining data exchange between ERP and SCADA systems.

SKILLS

Python, SQL, SCADA systems, ERP integration, LLMs, RAG, Model Evaluation, Experimentation, LangChain, Llama 3, Data pipelines, Data validation, Git, Docker,

LANGUAGES

English
IELTS: 8/9

German
B1-(actively improving)

SOCIAL NETWORKS

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